Quiz 7 — Solutions

M 375T Mathematical Foundations of Machine Learning — Fall 2026 · Thursday, October 22

Haar coordinates of $x = (2, 2, 2, 2, 0, 0, 6, 6)$ (10 points)

(a) Since the q_k are orthonormal, $c_k = q_k \cdot x$: $c_0 = \frac{8+12}{\sqrt{8}} = \frac{20}{\sqrt{8}} = 5\sqrt{2} \approx 7.07$, $\quad c_1 = \frac{8-12}{\sqrt{8}} = -\frac{4}{\sqrt{8}} = -\sqrt{2} \approx -1.41$, $\quad c_2 = \frac{1}{2}(2+2-2-2) = 0$, $\quad c_3 = \frac{1}{2}(0+0-6-6) = -6$, $\quad c_4 = c_5 = c_6 = c_7 = 0$. [5 pts]

(b) Onto q_0, q_1: $c_0 q_0 + c_1 q_1 = \frac{20}{8}(1, \ldots, 1) - \frac{4}{8}(1, 1, 1, 1, -1, -1, -1, -1) = (2, 2, 2, 2, 3, 3, 3, 3)$, the averages of x over the two halves. Error energy $= \sum_{k \geq 2} c_k^2 = c_3^2 = 36$ (directly: $\|(0, 0, 0, 0, -3, -3, 3, 3)\|^2 = 36$). Onto $q_0, \ldots, q_3$: add $c_3 q_3 = -6 \cdot \frac{1}{2}(0, 0, 0, 0, 1, 1, -1, -1) = (0, 0, 0, 0, -3, -3, 3, 3)$, giving $(2, 2, 2, 2, 0, 0, 6, 6) = x$. Error energy 0: all remaining coordinates are 0. [5 pts: 2 each projection, 1 for the two error energies]

Scoring: (a) 5, (b) 5. Partial credit for a correct method with an arithmetic slip. Mirrors Practice 7, B1.